HANNAH KIM, PH.D., M.S.

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WHO AM I?

- A postdoctoral research associate studying evolutionary genomics in cancer. Interested in understanding cancer phylogeny in the resolution of single-cell.
- Rooted in bioinformatics and hybridized (more appropriately, introgression given that it is a backcrossing) with human-computer interaction (HCI): Bioinformatics × HCI. Interested in bioinformatics education.
- A community builder who is good at initiating and facilitating collaborations.

EDUCATION

PhD in Bioinformatics, Temple University 08/2019 - 08/2026
Advisor, Bioinformatics: Dr. Sergei L Kosakovsky Pond
Co-Advisor, Human-Computer Interaction (HCI): Dr. Stephen MacNeil
Dissertation Topic: Harnessing interdisciplinarity in the investigation of nucleotide sequence evolution
[Recap | Abstract | Shorts | Dissertation]

MS in Computational Biology, Carnegie Mellon University 08/2015 - 12/2016
Relevant Coursework: Computational Genomics, Machine Learning, and Algorithms & Advanced Data Structures

BS in Chemistry, Carnegie Mellon University 08/2010 - 05/2013
Relevant Coursework: Principles of Computing, and Modern Analytical Instructions

MAJOR CERTIFICATES

08/2023 **Teaching in Higher Education Certificate** *Temple University - Center for the Advancement of Teaching*
08/2023 **MicroMBA** *University of California, San Diego Extended Studies - Rady School of Management*

SELECTED PUBLICATION

(4) Kim, H., Scheffler, K., Nekrutenko, A., Martin, D. P., Weaver, S., Murrell, B., Kosakovsky Pond, S. L. (2026). The Physicochemical Basis of Protein Evolution: Property-Informed Evolutionary Models (PRIME). *Molecular Biology and Evolution*, 1-49. doi:10.1093/molbev/msag200 [Preprint | Code | Web App | Data]

(3) Kim, H., Kosakovsky Pond, S. L., MacNeil, S. (2025). Conversations over Clicks: Impact of Chatbots on Information Search in Interdisciplinary Learning. *2025 IEEE Frontiers in Education Conference (FIE)*, 1-9. doi:10.1109/FIE63693.2025.11328556
- FIE 2025 [Preprint | Teasers | Slide | Video | Recap]

(2) Huzar, J., **Kim, H.**, Kumar, S., Miura, S. (2022). MOCA for integrated analysis of gene expression and genetic variation in single cells. *Frontiers in Genetics*, 13:831040. doi:10.3389/fgene.2022.831040

(1) Ding, Y., **Kim, H.**, Madden, K., Loftus, J., Chen, G., Allen, D., Zhang, R., Xu, J., Chen, C., Xu, Y., Tasian, S., Tan, K. (2021). Network Analysis Reveals Synergistic Genetic Dependencies for Rational Combination Therapy in Philadelphia Chromosome-like Acute Lymphoblastic Leukemia. *Clinical Cancer Research*, 27(18). doi:10.1158/1078-0432.CCR-21-0553

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate Sep 2026 - present
University of Mississippi – Sayaka Miura lab *Remote*

Project Lead Feb 2024 - May 2026
Temple University ACME Group & Temple University HCI lab *Philadelphia, PA*

- Identified gaps in knowledge and proposed ideas that can make data and method work together.
- Reconciled different research interests of stakeholders involved.
- Designed a user study, drafted a protocol approved by the institutional review board, and ran the study.

Bioinformatics Engineer Oct 2021 - Jan 2022
Lifetime Omics *Remote*

- Automated cutting-edge methods and analyzed COVID-19 metagenomics data in the Google Cloud environment.
- Developed detailed standard operating procedure for reproducibility.
- Took initiatives to solve problems and demonstrated dedication for the project in the startup environment.

Bioinformatics Analyst/Software Engineer Jul 2017 - Jun 2019
Children's Hospital of Philadelphia – Kai Tan lab *Philadelphia, PA*

- Investigated RNA-Seq and microarray data from B-Cell Acute Lymphoblastic Leukemia subtypes using differential gene expression analysis, Gene Ontology enrichment analysis, and other relevant bioinformatics methods.
- Identified cancer-specific genetic interactions that led to publications in high-impact journals.
- Facilitated communication in the interdisciplinary environment of doctors and wet-lab and dry-lab researchers.

TEACHING EXPERIENCE

Co-Coordinator/Reflective Practicum Completion for THE Certificate May 2023 - July 2023
Temple University *Deep Learning for the Life Sciences Journal Club*

- Co-organized a 10-week summer journal club with Dr. Enzo Carnevale for researchers of diverse levels of expertise (undergraduate students, graduate students, postdocs, alumni, visitors, and professors).
- Group size: 35.

Teaching Assistant Aug 2020 - Dec 2020
Temple University *Genomics in Medicine (BIOL-3111/5111)*

- Generated formative and summative assessment materials and provided timely feedback (Youtube/Introduction).
- Class size: 150.

Teaching Assistant Aug 2019 - May 2020
Temple University *Wet Lab Courses (BIOL-2112 and BIOL-1012)*

- Gave a short lecture in the beginning of every lab, monitored student performance, and provided guidance.
- Graded lab reports and generated quizzes.
- Class size: 20 (× 2 sections) each.
  - BIOL-1012 General Biology II, Spring 2020 was an introductory wet lab course for non-biology majors.
  - BIOL-2112 Introduction to Cellular and Molecular Biology, Fall 2019 was a lab for biology majors.

Course Developer Feb 2016 - Aug 2016
Carnegie Mellon University *Programming for Scientists (02201/02601)*

- Generated open-source course materials (codes and instructions) with Drs. Phillip Compeau and Carl Kingsford for Go-lang beginners.